Gavin Schmidt

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Technical Skills

Languages: C++, C#, HTML, CSS, JavaScript, Java, Python

Game Engines & Tools: Unreal Engine 5 (UE5), Unity 6, Visual Studio, VS Code, Git, GitHub

Core Competencies: Gameplay Frameworks, System Architecture, Object Pooling, AI Behavior Trees, Procedural Generation, Blueprint/C++ Interop, Technical Documentation & Production

Professional Experience

Gameplay Engineer Contractor — April 2025 – Present

Starchild PBC — *Orlando, FL(Remote)*

- Engineer core gameplay mechanics and systems in **C++** for an unannounced **mobile (iOS)** title in **Unreal Engine**.
- Architect robust inventory management and complex item interaction systems, ensuring scalable data flow and modularity.
- Optimize memory usage and data structures for mobile platform constraints while adhering to strict NDA and security guidelines.
- Collaborate directly with designers to iterate on moment-to-moment mechanics and implement responsive touch-based interactions.

Lead Gameplay Engineer & Producer — January 2025 – Present

UCF Capstone & Studio Teams (Teams: Mystical Die, Head or Tail, Splistream Studios, IceBreak Studios, Ghostknights, Tormentanine) — *Orlando, FL*

- Lead engineering and production efforts across **6 multi-disciplinary** academic studio teams, defining up to **95%** of the technical design surface area for multiple titles.
- Architect and implement core gameplay systems—including combat, grid-based movement, and dialogue—using **C++** and Unreal Blueprints.
- Engineer **A* pathfinding** algorithms, adaptive upgrade trees, and robust **binary serialization** systems for player data persistence.
- Expose underlying **C++** systems to designers via **Blueprint-extensible APIs**, cutting engineering dependency per new feature from full implementation to targeted review-only (**<35%**).
- Manage **sprint milestones**, conduct weekly **code reviews**, and author detailed **technical documentation** for **100+** scripts, significantly reducing regressions and designer implementation time.

Assistant Scholar (Gameplay Programmer) — May 2025 – July 2025

Limbitless Solutions LLC — *Orlando, FL*

- Provided product guidance across console/PC accessories and trade-ins
- Became familiar with gaming hardware and the effects that has on video game production and maintenance
- Handled **30–40** customer interactions/shift
- Performed inventory counts for **500+** SKUs with **>99% accuracy**

Assistant Scholar — May 2025 – July 2025

Limbitless Solutions LLC — *Orlando, FL*

- Built and integrated player-facing mechanics in **Unreal Engine** as part of a **25+** person interdisciplinary team, delivering changes on tight playtest deadlines.
- Translated playtest feedback from Bionic Kids into concrete gameplay changes (controls clarity, pacing, feedback/UI), improving moment-to-moment readability and engagement.
- Shipped weekly iteration builds and partnered with design to rapidly scope, implement, and validate changes in short development loops.

Technical Projects

Reconnection | *Unreal Engine 5, C++, Visual Studio, Git* — September 2025 – Present

- Engineered a **turn-based combat** game utilizing the **UE5 Gameplay Framework**, featuring a dialogue-responsive buff system and complex enemy AI driven by **Behavior Trees**.
- Designed and implemented combat mathematics inspired by Dungeons & Dragons' roll-to-hit and damage systems.
- Programmed intuitive player movement utilizing a combination of **Navmesh** and custom 2D grid mathematics.

Solar Scavengers | *Unity 6, C#, VS Code, Git* — January 2026

- Developed a PC game for the **FIEA Game Jam 2026** centered around momentum-based controls and resource management.
- Engineered a highly optimized **procedural generation** system utilizing **adaptive chunk-loading** and **object pooling**.
- Programmed dynamic game managers to scale difficulty, spawn rates, and **in-game lighting** responsively based on player progression.

Professional Experience

University of Central Florida — August 2021 – May 2026

Bachelor of Art in Digital Media – Game Design | Minor in Computer Science